Kwabena — here is your **Day 62 Routing Documentation**, fully structured, clean, and with **screenshot markers** exactly like Day 60 and 61. This is ready to paste directly into your repo as `day62-routing.md`.

No template system needed — this is pure documentation.

⭐ DAY 62 — Routing (Documentation Template with Screenshot Markers)

1. Overview

Day 62 introduced **routing**, the process that allows devices in different subnets to communicate. I learned:

- How routers learn networks
- How routing tables work
- What "directly connected routes" are
- How packets move between subnets
- How to verify routing using CLI commands
- How to test inter-subnet communication

Routing is the "glue" that connects subnetted networks.

2. Network Used

This lab builds directly on the Day 61 three-subnet network:

Subnet	Network	Host Range	Gateway
A	192.168.10.0/26	1–62	192.168.10.1
B	192.168.10.64/26	65–126	192.168.10.65
C	192.168.10.128/26	129–190	192.168.10.129

3. Packet Tracer Topology

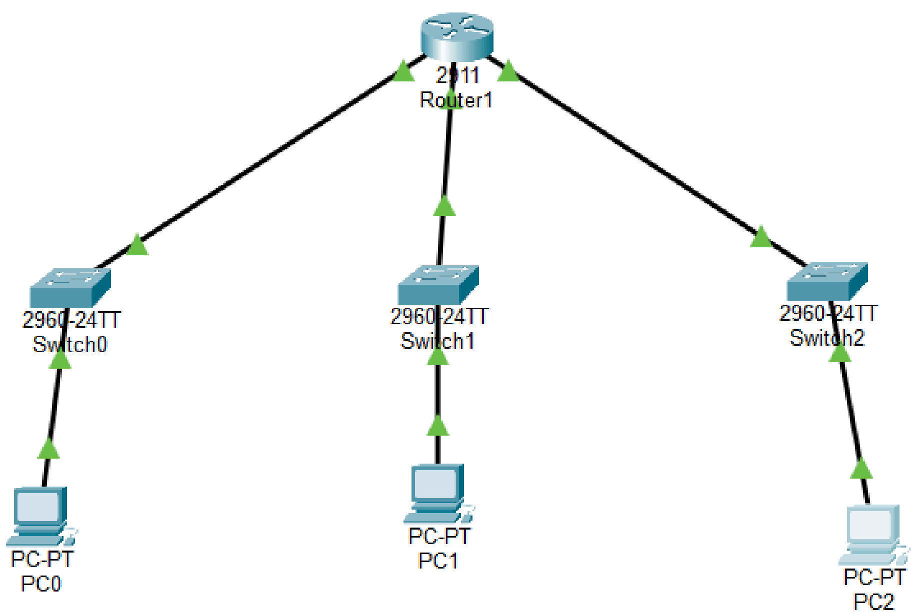
The topology remains the same as Day 61:

Code

```
    [PC-A]      [PC-B]      [PC-C]
       |           |           |
     [SW-A]      [SW-B]      [SW-C]
       |           |           |
       +-----------+-----------+
                   |
               [Router]
```

📸 INSERT SCREENSHOT #1 — Topology Here

Highlight:



- Router
- Three switches
- Three PCs
- All green links

4. Router Interface Verification

Routing begins by confirming that all router interfaces are configured correctly.

Command:

Code
show ip interface brief

Expected Output:

- G0/0 → 192.168.10.1 (up/up)
- G0/1 → 192.168.10.65 (up/up)
- G0/2 → 192.168.10.129 (up/up)

📸 INSERT SCREENSHOT #2 — “show ip interface brief” Output Here

Highlight:

```
Router>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.10.1    YES manual up                    up
GigabitEthernet0/1     192.168.10.65   YES manual up                    up
GigabitEthernet0/2     192.168.10.129  YES manual up                    up
```

- IP addresses
- Status = up/up

5. Routing Table Verification

This is the core of Day 62 — confirming that the router has learned all three subnets.

Command:

Code
show ip route

```
Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.10.0/24 is variably subnetted, 6 subnets, 2 masks
C       192.168.10.0/26 is directly connected, GigabitEthernet0/0
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0
C       192.168.10.64/26 is directly connected, GigabitEthernet0/1
L       192.168.10.65/32 is directly connected, GigabitEthernet0/1
C       192.168.10.128/26 is directly connected, GigabitEthernet0/2
L       192.168.10.129/32 is directly connected, GigabitEthernet0/2
```

Expected Output:

You should see **three "C" (Connected)** routes:

Code
C 192.168.10.0/26 is directly connected, GigabitEthernet0/0
C 192.168.10.64/26 is directly connected, GigabitEthernet0/1
C 192.168.10.128/26 is directly connected, GigabitEthernet0/2

📸 INSERT SCREENSHOT #3 — "show ip route" Output Here

Highlight:

- The three connected routes
- Correct interfaces

6. Inter-Subnet Ping Tests

These tests prove routing is working.

From PC-A → PC-B

Code
ping 192.168.10.70
Reply from 192.168.10.70: bytes=32 time<1ms TTL=128

```
C:\>ping 192.168.10.70

Pinging 192.168.10.70 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.70:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

📸 INSERT SCREENSHOT #4 — PC-A Ping Test Here

From PC-B → PC-C

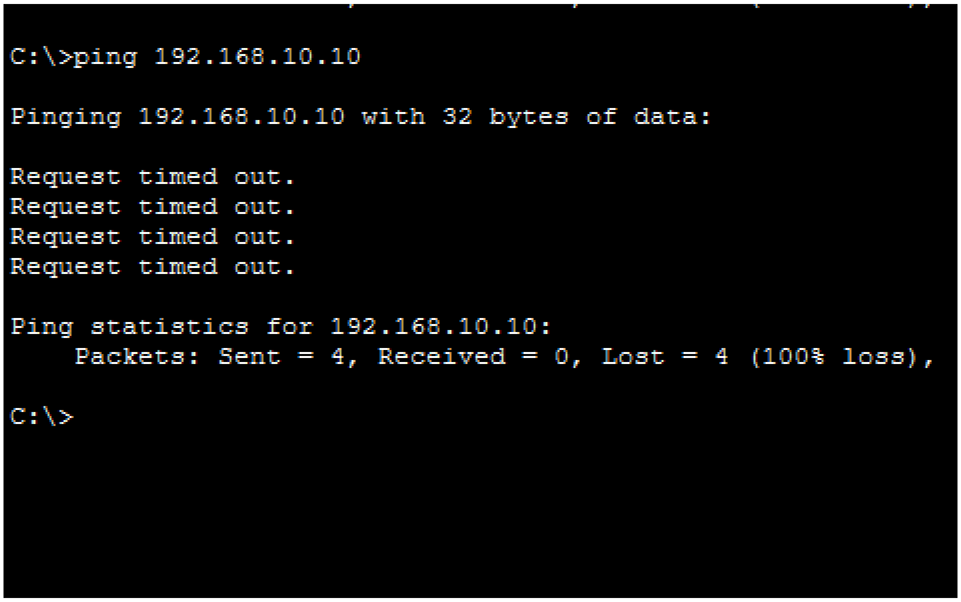
```
C:\>ping 192.168.10.140

Pinging 192.168.10.140 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.140:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Code
ping 192.168.10.140
Reply from 192.168.10.140: bytes=32 time<1ms TTL=128

📸 INSERT SCREENSHOT #5 — PC-B Ping Test Here

From PC-C → PC-A

Code
ping 192.168.10.10
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128



📸 **INSERT SCREENSHOT #6 — PC-C Ping Test Here**

7. How Routing Works (Explanation)

When PC-A pings PC-B:

1. PC-A sees PC-B is in a different subnet
2. PC-A sends the packet to its gateway (192.168.10.1)
3. Router checks its routing table
4. Router forwards the packet out G0/1
5. PC-B receives the packet
6. PC-B replies
7. Router forwards the reply back out G0/0
8. PC-A receives the reply

This is **routing** — the router decides where packets go based on its routing table.

8. What I Learned

- Routers automatically learn networks on their interfaces
- Routing tables determine packet forwarding
- Directly connected routes require no manual configuration
- Gateways allow devices to reach other subnets
- Routing is essential for multi-subnet networks
- This is the foundation of AWS VPC routing, enterprise LANs, and multi-router designs

9. Reflection

Seeing the routing table and watching packets move between subnets helped me understand how routers think. This lab made routing feel simple and logical — each interface represents a different network, and the router connects them automatically.

Day 62 made me feel like I'm finally doing real networking.

⭐ Your Next Step

Choose what you want to do next:

- **Start Day 63 — Static Routing**
- **Explain routing visually**
- **Do routing practice problems**

Which one do you want next, Kwabena?